COHOMOLOGICAL DESCENT FOR THE PH TOPOLOGY: PI_X^{-1}K = R EPSILON_{X,*} A_X^{-1}K

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-cohomological-descent-etale-ph (source lines 1962-1973); the proof environment beginning on the line after the following blank line (source line 1975) is its own proof and closes at source line 2005.

Editorial note. The proof invokes two results of the same chapter by label, both retained as context with the generator's transitive expansion (note that lemma-comparison-ph-etale is itself a counted card of this lane on the disjoint interval 1888-1955); four cross-chapter results are each declared as a tagged source pointer.

Editorial note. The author proves only the second statement and says the first is easier and proved in exactly the same manner; that remark is reproduced verbatim and the card supplies nothing in its place.

Editorial note. The retained preamble is the chapter's own (sources/stacks-spaces-more-cohomology/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup and theorem declarations; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's chapter Cohomology of Algebraic Spaces, counted once: cohomological descent for the ph topology, i.e. pi_X^{-1}K = R epsilon_{X,*}a_X^{-1}K and K = R a_{X,*}a_X^{-1}K for torsion complexes.

Allowed standard prerequisites (editorial).

- Torsion complexes and their reduction to sheaves. A complex K in D^+(X_etale) with torsion cohomology sheaves can be represented by a bounded below complex of torsion abelian sheaves (Cohomology on Sites, declared as a tagged source pointer).. It is the reduction at source lines 1977-1981.
- Leray's acyclicity lemma. If a complex is acyclic for the higher direct images of the functor, then the higher direct images of the complex vanish in positive degrees (Derived Categories, declared as a tagged source pointer).. It finishes the complex case at source lines 1985-1988.
- The comparison lemma for the ph topology. a_{X,*}a_X^{-1}F = F and R^i epsilon_{X,*}(a_X^{-1}F) = 0 for i > 0 for torsion F (the author's lemma-comparison-ph-etale, retained as context).. It gives the identity and one vanishing at source lines 1992-1999.

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- Descent along the projection to the small etale site. R^j pi_{X,*}(pi_X^{-1}F) = 0 for j > 0 (the author's lemma-cohomological-descent-etale, retained as context).. It supplies the other vanishing in the Leray decomposition at source lines 2002-2003.
- The relative Leray spectral sequence. The Leray spectral sequence for a composite functor, used for a_X = epsilon_X o pi_X (Cohomology on Sites, declared as a tagged source pointer).. It combines the two vanishing statements at source lines 1995-1997.

Source reference. Original lemma 0DD7: sources/stacks-chow/tags.txt, source lines 16757–16757. Cross-chapter pointer whose target carries the canonical tag 0DD7 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1505–1505. Cross-chapter pointer whose target carries the canonical tag 015E in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0734: sources/stacks-chow/tags.txt, source lines 9048–9048. Cross-chapter pointer whose target carries the canonical tag 0734 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DE4: sources/stacks-chow/tags.txt, source lines 16789–16789. Cross-chapter pointer whose target carries the canonical tag 0DE4 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 072W: sources/stacks-chow/tags.txt, source lines 9040–9040. Cross-chapter pointer whose target carries the canonical tag 072W in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03F3: sources/stacks-chow/tags.txt, source lines 4478–4478. Cross-chapter pointer whose target carries the canonical tag 03F3 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DDW: sources/stacks-chow/tags.txt, source lines 16781–16781. Cross-chapter pointer whose target carries the canonical tag 0DDW in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03CG: sources/stacks-chow/tags.txt, source lines 4383–4383. Cross-chapter pointer whose target carries the canonical tag 03CG in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03A0: sources/stacks-chow/tags.txt, source lines 4291–4291. Cross-chapter pointer whose target carries the canonical tag 03A0 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DFM: sources/stacks-chow/tags.txt, source lines 16842–16842. Cross-chapter pointer whose target carries the canonical tag 0DFM in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DFH: sources/stacks-chow/tags.txt, source lines 16837–16837. Cross-chapter pointer whose target carries the canonical tag 0DFH in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 09XN: sources/stacks-chow/tags.txt, source lines 12536–12536. Cross-chapter pointer whose target carries the canonical tag 09XN in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 04K2: sources/stacks-chow/tags.txt, source lines 5958–5958. Cross-chapter pointer whose target carries the canonical tag 04K2 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 089J: sources/stacks-chow/tags.txt, source lines 10499–10499. Cross-chapter pointer whose target carries the canonical tag 089J in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DES: sources/stacks-chow/tags.txt, source lines 16812–16812. Cross-chapter pointer whose target carries the canonical tag 0DES in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S.*

1. *For $\mathcal{F} \in Sh(X_{étale})$ we have $\epsilon_{X,*}a_X^{-1}\mathcal{F} = \pi_X^{-1}\mathcal{F}$ and $a_{X,*}a_X^{-1}\mathcal{F} = \mathcal{F}$.*
2. *For $\mathcal{F} \in Ab(X_{étale})$ torsion we have $R^i\epsilon_{X,*}(a_X^{-1}\mathcal{F}) = 0$ for $i > 0$.*

Proof. We have $a_X^{-1}\mathcal{F} = \epsilon_X^{-1}\pi_X^{-1}\mathcal{F}$. By Lemma 0.4 the étale sheaf $\pi_X^{-1}\mathcal{F}$ is a sheaf for the ph topology and therefore is equal to $a_X^{-1}\mathcal{F}$ (as pulling back by ϵ_X is given by ph sheafification). Recall moreover that $\epsilon_{X,*}$ is the identity on underlying presheaves. Now part (1) is immediate from the explicit description of π_X^{-1} in Lemma 0.4.

We will prove part (2) by reducing it to the case of schemes – see part (1) of Étale Cohomology, Lemma 0DE4. This will "clearly work" as every algebraic space is étale locally a scheme. The details are given below but we urge the reader to skip the proof.

For an abelian sheaf $\mathcal{H}$ on $(Spaces/X)_{ph}$ the higher direct image $R^p\epsilon_{X,*}\mathcal{H}$ is the sheaf associated to the presheaf $U \mapsto H^p_{ph}(U, \mathcal{H})$ on $(Spaces/X)_{étale}$. See Cohomology on Sites, Lemma 072W. Since every object of $(Spaces/X)_{étale}$ has a covering by schemes, it suffices to prove that given U/X a scheme and $\xi \in H^p_{ph}(U, a_X^{-1}\mathcal{F})$ we can find an étale covering $\{U_i \to U\}$ such that ξ restricts to zero on U_i. We have

$$\begin{aligned} H^p_{ph}(U, a_X^{-1}\mathcal{F}) &= H^p((Spaces/U)_{ph}, (a_X^{-1}\mathcal{F})|_{Spaces/U}) \\ &= H^p((Sch/U)_{ph}, (a_X^{-1}\mathcal{F})|_{Sch/U}) \end{aligned}$$

where the second identification is Lemma 0.3 and the first is a general fact about restriction (Cohomology on Sites, Lemma 03F3). Looking at the first paragraph and the corresponding result in the case of schemes (Étale Cohomology, Lemma 0DDW) we conclude that the sheaf $(a_X^{-1}\mathcal{F})|_{Sch/U}$ matches the pullback by the "schemes version of a_U". Therefore we can find an étale covering $\{U_i \to U\}$ such that our class dies in $H^p((Sch/U_i)_{ph}, (a_X^{-1}\mathcal{F})|_{Sch/U_i})$ for each i, see Étale Cohomology, Lemma 0DE4 (the precise statement one should use here is that V_n holds for all n which is the statement of part (2) for the case of schemes). Transporting back (using the same formulas as above but now for U_i) we conclude ξ restricts to zero over U_i as desired. □

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. For $K \in D(X_{étale})$ the map*

$$K \longrightarrow R\pi_{X,*}\pi_X^{-1}K$$

is an isomorphism where $\pi_X : Sh((Spaces/X)_{étale}) \to Sh(X_{étale})$ is as above.

Proof. This is true because both π_X^{-1} and $\pi_{X,*} = i_X^{-1}$ are exact functors and the composition $\pi_{X,*} \circ \pi_X^{-1}$ is the identity functor. □

Lemma 0.3. *Let S be a scheme. Let $\tau \in \{étale, fppf, ph\}$ (add more here). The inclusion functor*

$$(Sch/S)_\tau \longrightarrow (Spaces/S)_\tau$$

is a special cocontinuous functor (Sites, Definition 03CG) and hence identifies topoi.

Proof. The conditions of Sites, Lemma 03A0 are immediately verified as our functor is fully faithful and as every algebraic space has an étale covering by schemes. □

Lemma 0.4. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\pi_X^{-1}\mathcal{F}$ is given by the rule*

$$(\pi_X^{-1}\mathcal{F})(Y) = \Gamma(Y_{étale}, f_{small}^{-1}\mathcal{F})$$

for $f : Y \to X$ in $(Spaces/X)_{étale}$. Moreover, $\pi_Y^{-1}\mathcal{F}$ satisfies the sheaf condition with respect to smooth, syntomic, fppf, fpqc, and ph coverings.

Proof. Since pullback is transitive and $f_{small} = \pi_X \circ i_f$ (see above) we see that $i_f^{-1}\pi_X^{-1}\mathcal{F} = f_{small}^{-1}\mathcal{F}$. This shows that π_X^{-1} has the description given in the lemma.

To prove that $\pi_X^{-1}\mathcal{F}$ is a sheaf for the ph topology it suffices by Topologies on Spaces, Lemma 0DFM to show that for a surjective proper morphism $V \to U$ of algebraic spaces over X we have $(\pi_X^{-1}\mathcal{F})(U)$ is the equalizer of the two maps $(\pi_X^{-1}\mathcal{F})(V) \to (\pi_X^{-1}\mathcal{F})(V \times_U V)$. This we have seen in Lemma 0.5.

The case of smooth, syntomic, fppf coverings follows from the case of ph coverings by Topologies on Spaces, Lemma 0DFH.

Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be an fpqc covering of algebraic spaces over X. Let $s_i \in (\pi_X^{-1}\mathcal{F})(U_i)$ be sections which agree over $U_i \times_U U_j$. We have to prove there exists a unique $s \in (\pi_X^{-1}\mathcal{F})(U)$ restricting to s_i over U_i. Case I: U and U_i are schemes. This case follows from Étale Cohomology, Lemma 09XN. Case II: U is a scheme. Here we choose surjective étale morphisms $T_i \to U_i$ where T_i is a scheme. Then $\mathcal{T} = \{T_i \to U\}$ is an fpqc covering by schemes and by case I the result holds for $\mathcal{T}$. We omit the verification that this implies the result for $\mathcal{U}$. Case III: general

case. Let $W \to U$ be a surjective étale morphism, where W is a scheme. Then $\mathcal{W} = \{U_i \times_U W \to W\}$ is an fpqc covering (by algebraic spaces) of the scheme W. By case II the result hold for $\mathcal{W}$. We omit the verification that this implies the result for $\mathcal{U}$. □

Lemma 0.5. *Let S be a scheme. Let $f : Y \to X$ be a surjective proper morphism of algebraic spaces over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective with image the equalizer of the two maps $f_*f^{-1}\mathcal{F} \to g_*g^{-1}\mathcal{F}$ where g is the structure morphism $g : Y \times_X Y \to X$.*

Proof. For any surjective morphism $f : Y \to X$ of algebraic spaces over S, the map $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective. Namely, if $\overline{x}$ is a geometric point of X, then we choose a geometric point $\overline{y}$ of Y lying over $\overline{x}$ and we consider

$$\mathcal{F}_{\overline{x}} \to (f_*f^{-1}\mathcal{F})_{\overline{x}} \to (f^{-1}\mathcal{F})_{\overline{y}} = \mathcal{F}_{\overline{x}}$$

See Properties of Spaces, Lemma 04K2 for the last equality.

The second statement is local on X in the étale topology, hence we may and do assume Y is an affine scheme.

Choose a surjective proper morphism $Z \to Y$ where Z is a scheme, see Cohomology of Spaces, Lemma 089J. The result for $Z \to X$ implies the result for $Y \to X$. Since $Z \to X$ is a surjective proper morphism of schemes and hence a ph covering (Topologies, Lemma 0DES) the result for $Z \to X$ follows from Étale Cohomology, Lemma 0DDW (in fact it is in some sense equivalent to this lemma). □

Lemma 0.6. *Let S be a scheme. Let X be an algebraic space over S. For $K \in D^+(X_{étale})$ with torsion cohomology sheaves the maps*

$$\pi_X^{-1}K \longrightarrow R\epsilon_{X,*}a_X^{-1}K \quad \text{and} \quad K \longrightarrow Ra_{X,*}a_X^{-1}K$$

are isomorphisms with $a_X : Sh((Spaces/X)_{ph}) \to Sh(X_{étale})$ as above.

Proof. We only prove the second statement; the first is easier and proved in exactly the same manner. There is a reduction to the case where K is given by a single torsion abelian sheaf. Namely, represent K by a bounded below complex $\mathcal{F}^\bullet$ of torsion abelian sheaves. This is possible by Cohomology on Sites, Lemma 0DD7. By the case of a sheaf we see that $\mathcal{F}^n = a_{X,*}a_X^{-1}\mathcal{F}^n$ and that the sheaves $R^qa_{X,*}a_X^{-1}\mathcal{F}^n$ are zero for $q > 0$. By Leray's acyclicity lemma (Derived Categories, Lemma 015E) applied to $a_X^{-1}\mathcal{F}^\bullet$ and the functor $a_{X,*}$ we conclude. From now on assume $K = \mathcal{F}$ where $\mathcal{F}$ is a torsion abelian sheaf.

By Lemma 0.1 we have $a_{X,*}a_X^{-1}\mathcal{F} = \mathcal{F}$. Thus it suffices to show that $R^qa_{X,*}a_X^{-1}\mathcal{F} = 0$ for $q > 0$. For this we can use $a_X = \epsilon_X \circ \pi_X$ and the Leray spectral sequence (Cohomology on Sites, Lemma 0734). By Lemma 0.1 we have $R^i\epsilon_{X,*}(a_X^{-1}\mathcal{F}) = 0$ for $i > 0$. We have $\epsilon_{X,*}a_X^{-1}\mathcal{F} = \pi_X^{-1}\mathcal{F}$ and by Lemma 0.2 we have $R^j\pi_{X,*}(\pi_X^{-1}\mathcal{F}) = 0$ for $j > 0$. This concludes the proof. □